



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP 1513

TECHNOLOGY AND INFORMATION SYSTEM

Section : 02

Lecturer:

Dr Aryati binti Bakri

Digital Thinking Group Assignment
[Artificial Intelligence]

Group 4 Members:

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2	BRENDAN CHIA YAN FEI	A23CS0211
3	NEO LI XIN	A23CS0253
4	LEE YIN SHEN	A23CS0236

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Interview question preparation

Introduction

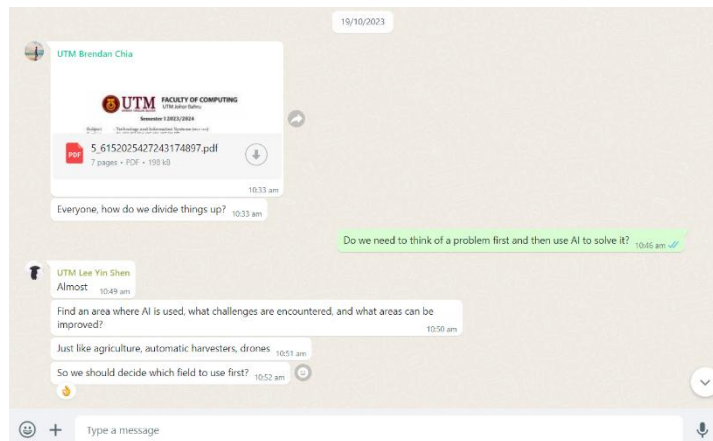
Design thinking is a non-linear, iterative process that people use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test. In short, design thinking is a philosophy and a set of tools to help you solve problems creatively. “Design thinking draws on logic, imagination, intuition, and systemic reasoning to explore the possibilities of what could be, and to create desired outcomes that benefit the end user, which is the customer. A design mind-set is not problem focused, it’s solution-focused, and action-oriented. It involves both analysis and imagination.” This is quoted by Linda Naiman, the CEO of Creativity at Work. There are five phases of design thinking, empathise, define, ideate, prototype and test.

Project Timeline

[22 October 2023]

Project Kick-off & First Meeting

[Empathy]

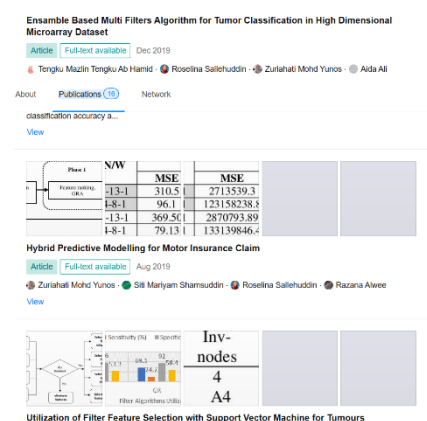


In our team's first meeting, we created a WhatsApp group for future discussion. We have also discussed and selected our group topic. We have reached mutual agreement on selecting the topic of Artificial Intelligence (AI) as our group's main topic. By selecting the topic of AI, we have come out with a list of potential lecturers with area of expertise in artificial intelligence to have an interview session with them. We then drafted initial interview questions to be conducted during the interview session.

[2 November 2023]

Request Permission from Expert

[Empathy]

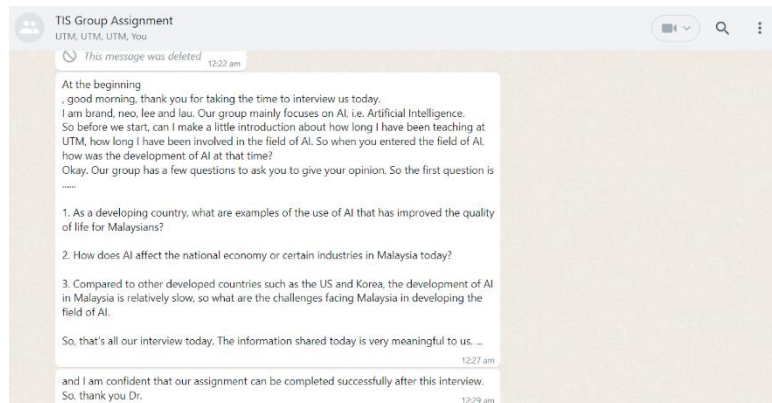


We have contacted potential lecturers with their areas of expertise in the field of AI and requested their participation via the WhatsApp platform. We have confirmed their availability and schedule for an interview session. To confirm the lecturers' area of expertise, we have reviewed their background carefully through UTM Scholar, and Research Gate website.

[5 November 2023]

Interview Question Preparation

[Ideate]

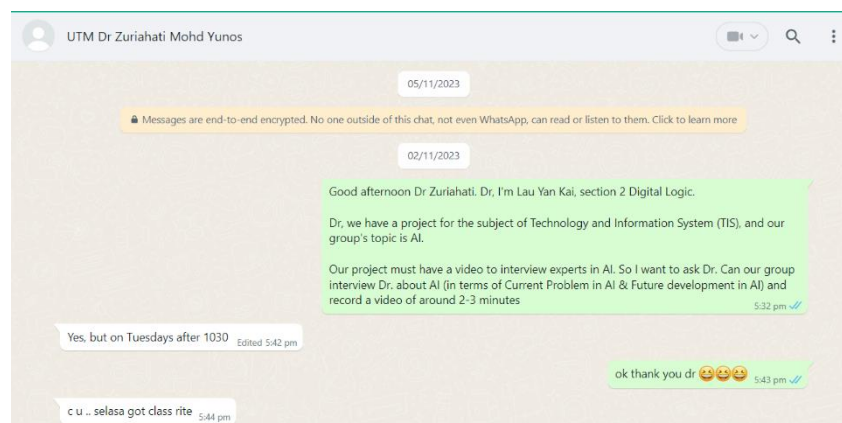


We have a discussion on finalising the interview question. We did search online to get more information regarding the topic before preparation of the interview script. After finalising the interview questions and receiving feedback from each team member, the first interview script draft was written. The interview script draft was shared with the team members via WhatsApp platform.

[5 November 2023]

Before-Interview Preparation

[Empathy]



We have communicated with the lecturer to confirm and double check, as well as a reminder, her availability in the interview session. We discussed the date and time to have an interview, as well as the venue where it is convenient for both parties.

[5 November 2023]

Second Meeting with Teammates

[Define]



We have a short online video conferencing session to discuss the responsibilities and roles of the team member, as well as expectations of the interview. Before the interview session, we checked and reviewed our schedule so that the interview session would not crash with any classes being carried out.

[7 November 2023]

Interview with Expert

[Empathy]



We conducted a face-to-face interview with Dr Zuriahati Mohd Yunus, a respected lecturer specialising in the field of artificial intelligence. The interview took place at BK7, N28, Faculty of Computing. The entire interview session was recorded to capture the opinions shared by Dr Zuriahati. The key points conveyed by Dr Zuriahati were noted for future reference and report writing.

[8 November 2023]

Documentation after Interview

[Define]



Following the interview session with Dr Zuriahati Mohd Yunos, we had a meeting to discuss on the documentation tasks. The priority of the meeting was to transcribe the interview and note down the key points conveyed by the expert. Amongst the information extracted from the interview, the attention was given to identifying and understanding the main current problems in the field of artificial intelligence and measures to solve them. We did also stress about the application of artificial intelligence in our real-life world.

[13 November 2023]

Brainstorming and Prototyping

[Prototype]

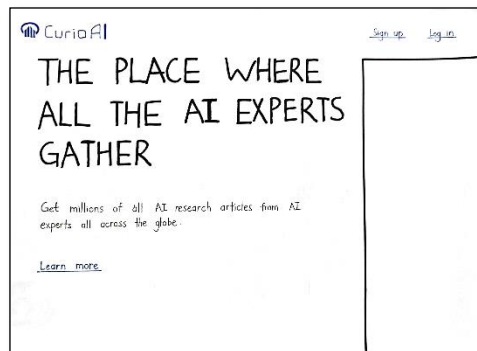


We have a collaborative brainstorming session to identify and think of the ideas for the application of artificial intelligence to our surroundings. From the questions and answered exchanged during the interview with Dr Zuriahati Mohd Yunos, we recognised the concept, current problems, and its solutions. We applied the opinion conveyed by Dr Zuriahati and turned it into a prototype.

[15 November 2023]

Prototype Model Testing

[Test]

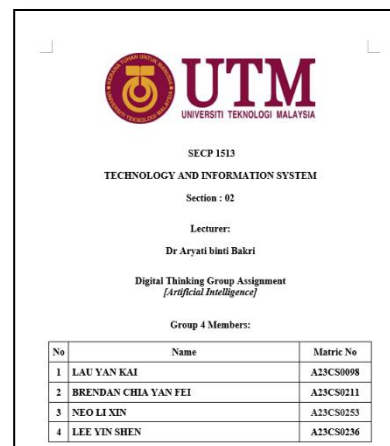
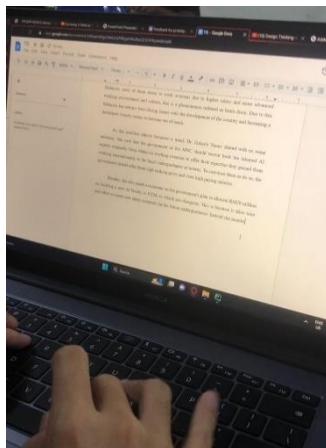


We have finished designing and creating of our initial prototype, and we were evaluating the effectiveness of the prototype during this phase. We evaluated the effectiveness, usability, and its ability to solve the main current problems identified in the field of artificial intelligence as mentioned in the interview with Dr Zuriahati Mohd Yunos. At the same time, we identified the potential challenges within the field of artificial intelligence. After the first testing of prototype model, we make adjustments towards the prototype model in order to enhance the overall efficiency of the prototype.

[15 November 2023]

Preparation for Report Writing

[Define]



After interviewing the expert and making the prototype, we completed our formal writing report. This report serves as a formal reference for the lecturer to check our group progress and evidence of conductance of the group project. The purpose of this report is also to record the information obtained from various of resources.

[15 November 2023]

Reflection on the Project

[Define]

Brendan Chia Yan Fei



My goal is to gain experience in real-world applications. Work-based learning projects allows me to apply my knowledge in real-world situations. Besides, I hope to increase my networking opportunities through this program. I will build a strong foundation by mastering the basics of programming skills. Besides, I will focus on the quality of my assignments and work-based learning projects to have a well-built portfolio.

Lee Yin Shen



Hopefully I can learn and remember all the syllabus and can explain my knowledge to those who are wondering the technology and information system. This design thinking process helps me to learn and research something systematically. I will try to use this approach for my further study. I will attend more programs and industry talks to fulfil myself with knowledge. Moreover, I will keep focus on my study and enhance my soft skill.

Lau Yau Kai



As a data engineering student at UTM, my goal is to become a skilled and innovative Data Engineer in the future. I hope to apply what I did learn during my studies and develop solutions for real-life problems and challenges. Design thinking influences my goal in thinking solutions and becoming a creative problem-solver. It allows me to learn and refine solutions creatively while collaborating with others. To improve my potential in industry, I'll master programming skills, attend workshops, participate in hackathon, and become an outstanding student amongst others.

Neo Li Xin



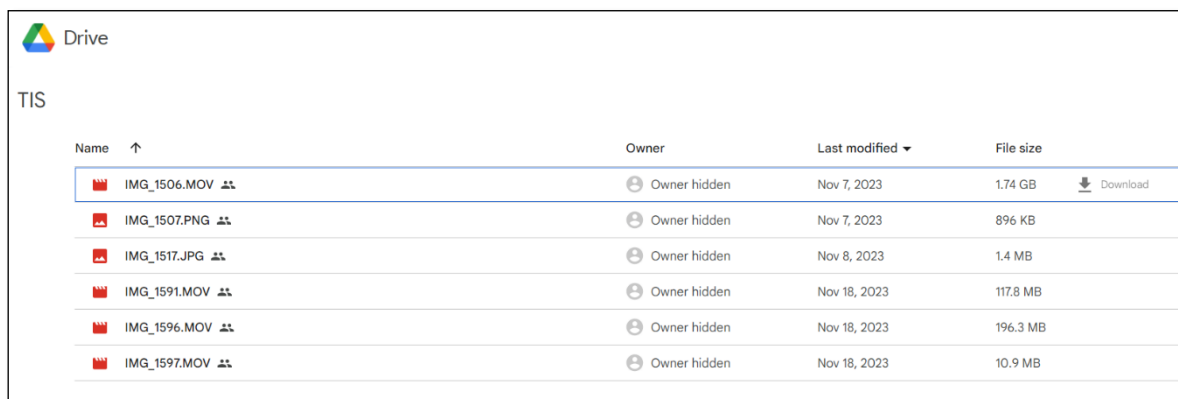
I wish to enhance and improve my computer knowledge so that I could explain to others simple enough for anyone to understand. It trains me to think logically and critically to solve any problems that I'll face in the future. Not to mention I get to refine my video editing skills. I'll join more outdoor activities or programs to boost my communication and leadership skills. Besides, I'll practice my programming skills regularly.

After completing the report and interviewing with the AI expert, we have almost come to an end of the project. We identified areas of improvement that can be implemented to improve ourselves in the industry. In addition, we highlighted our goals after graduation with regard to the course. Apart from that, we identified the outcome and impact of design thinking with regard to data engineering course.

[16 November 2023]

Preparation for Video Recording

[Ideate]



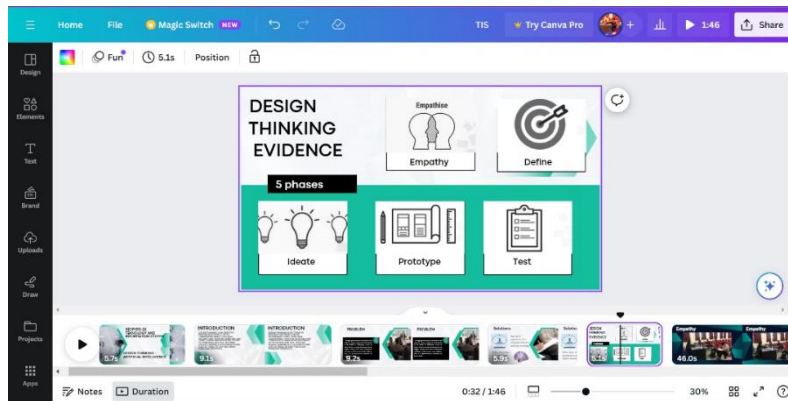
Name	Owner	Last modified	File size
IMG_1506.MOV	Owner hidden	Nov 7, 2023	1.74 GB
IMG_1507.PNG	Owner hidden	Nov 7, 2023	896 KB
IMG_1517.JPG	Owner hidden	Nov 8, 2023	1.4 MB
IMG_1591.MOV	Owner hidden	Nov 18, 2023	117.8 MB
IMG_1596.MOV	Owner hidden	Nov 18, 2023	196.3 MB
IMG_1597.MOV	Owner hidden	Nov 18, 2023	10.9 MB

Video recordings were checked and made sure that the video recordings captured meet the standards. The video recordings quality was clear, the voices are clear and easy to listen to, and the overall brightness of the video recordings were ensured. Simultaneously, we prepared script for the video recordings. The script serves as a guideline for our recordings, editing process and help the recording process carry out smoothly.

[17 November 2023]

Video Editing

[Ideate]



We edited the video recordings using video editing website, Canva. Advanced noise reduction was implemented in the video in order to enhance the voice quality in the video. The editing process also involves the removal of unnecessary parts within the videos. The step help reduce the length of the video, making it short and concise by maximum of 5 minutes. Besides, we also added visual elements in the video such as seamless transitions between scenarios. The output edited video was in high definition and uploaded to YouTube platform.

[17 November 2023]

Project Finalisation

[Ideate]

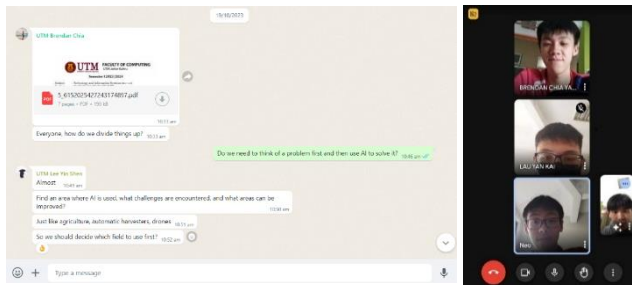


In the last and most important phase of project finalisation, we reviewed our project again to ensure that the project aligns with the standards and requirements. The report and video was carefully reviewed, the report was checked for its accuracy, and the video was checked for its overall presentation results. Ultimately, last-minute adjustments are made to any little errors or inconsistencies identified during the review.

Design Thinking Phases

[Empathy]

▪ Discussion on our topic



In our first meeting, we made a WhatsApp group for ongoing discussions and agreed on Artificial Intelligence (AI) as our main topic.

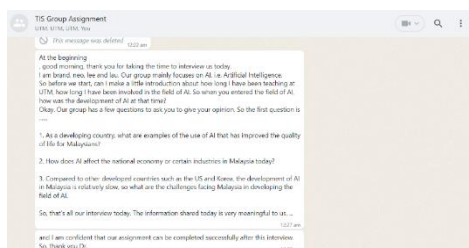
▪ Preparation for interview with expert



We have contacted Dr Zuriahati Mohd Yunus, who has more than 15 publications related to the topic of AI with more than 75 citations. We have confirmed their availability and schedule

for an interview session. We hope to obtain valuable insight from her.

▪ Preparation for interview question



We have a discussion on finalising the interview question. The finalised interview questions are:

1. Based on your experience, what are the challenging problems you have encountered in the field of AI?
2. Recently, Prime Minister Anwar announced an initial

allocation of RM20 million to establish Malaysia's first AI faculty at UTM. Do you think how significant is this funding in shaping the future of AI in Malaysia, and what potential breakthroughs do you foresee in this context?

▪ Interview with expert



We conducted a face-to-face interview with Dr Zuriahati Mohd Yunus. The interview took place at BK7, N28, Faculty of Computing. Dr Zuriahati had shared us a lot of valuable insight of hers and the potential solutions to solve the current problem of AI in Malaysia.

[Define]

Malaysia, as a developing country, compared to other developed countries such as the United States of America or the United Kingdom, has encountered significant challenges in the field of Artificial Intelligence (AI). According to valuable insights from Dr Zuriahati, an experienced lecturer, with more than 15 publications published related to the topic of AI, at UTM, the greatest challenge in AI development in Malaysia is the lack of skilled professionals in the field. The mentioned issue not only slower down the development progress of AI, but also brings tangible effects to the quality of life from all walks of life of the citizen of Malaysia. As a result, there are only a few approaches that Malaysian authorities, including the government, social media, public, or NGOs need to make the effect in order to promote the development of AI in our country. It is a fact that, even though education was able to create talented individuals in AI field in Malaysia, most of them chose to work abroad due to higher pay and more advanced working environment and culture. The scenario can be referred to as a phenomenon called brain drain. Due to the reason, Malaysia has always been facing issues with the development of the country and becoming a developed country seems to become out of reach.

[Ideate]

To overcome the challenge of shortage of AI experts, the Malaysian government should implement approaches to recruit back the AI experts who currently working overseas. In such scenario, the Malaysian government could pay competitive salary, giving great titles and posts, and establish attractive employment contract to those experts to overcome the issue of brain drain. The strategy could encourage those experts bring back their professional skills and experiences, as well as their knowledge learnt overseas, to contribute to our local AI community.

Besides, the Malaysian government should also focus, and improve on the faculty of AI at Universiti Malaya (UM). Since UM already has their faculty of AI, the government should focus on improving the faculty, instead of allocating another budget and propose an idea of allocating RM20 millions to establish another Faculty of AI in Universiti Teknologi Malaysia (UTM).

Apart from that, we have come to an idea which we could build up an online learning platform, initially as a website. The main objective of the website is to collect articles, journal or research papers related to the topic of Artificial Intelligence or Machine Learning by utilising an automated algorithm. As the founders of the learning platform, plan to propose and have partnership with international experts in AI from different regions, such as the United States of America, the United Kingdom, and Taiwan. Inviting the experts for livestreaming sessions, we aim to foster interaction and knowledge sharing with interested viewers. Additionally, we will also have Memorandum of Understanding (MoU) with local universities, to provide continuous knowledge and support to undergraduate, as well as postgraduate students, who currently studying in the field of Computer Science or Artificial Intelligence.

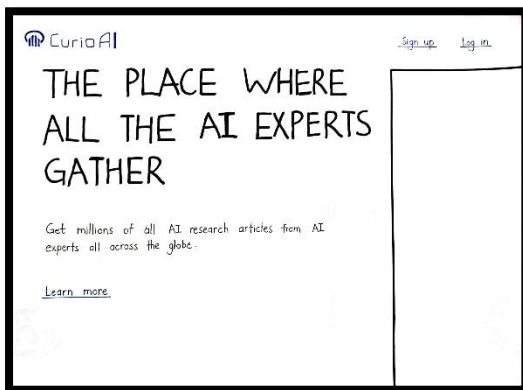
[Prototype]

After group discussions and brainstorming, we have generated an idea which we could build up an online learning platform. The website will be an online learning platform dedicated to Artificial Intelligence (AI) and Machine Learning. Its primary function is to automatically collect the valuable articles, journal articles, and research papers related to these topics based on an advanced algorithm. The website is accessible to the public. The students and professionals who would like to have an in-depth insight into the realm of AI could access the website easily and obtain information from the website.

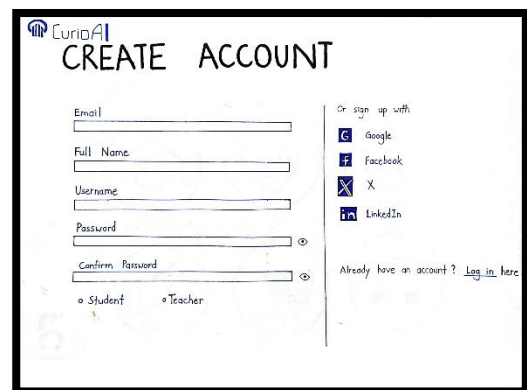
The website serves as a hub that promotes continuous learning and knowledge exchange amongst the AI community. Through partnerships with international AI experts from various of regions, livestreaming sessions, the platform helps cultivate more AI expert in our country by facilitating interactive learning experiences and advancing their understanding about AI and its applications.

Through Memorandum of Understanding (MoU) with local universities, the platform extends its impact by collaborating with academic institutions. The collaboration with local universities helps provide continuous support and educational resources to undergraduate and postgraduate students who currently pursuing their studies in Computer Science or AI. In such situations, our platform would help contribute to the growth and development of AI in our country.

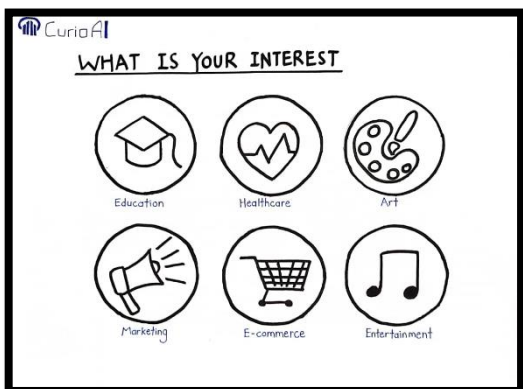
[Prototype Testing]



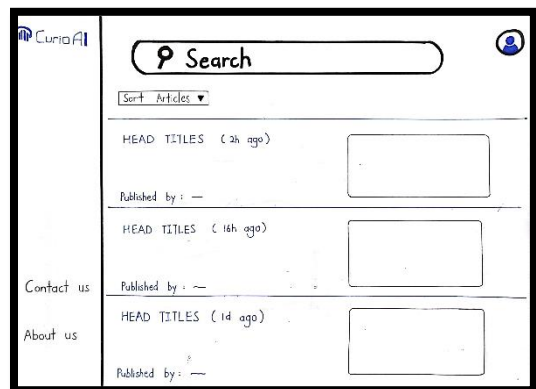
Main page of our website -CurioAI



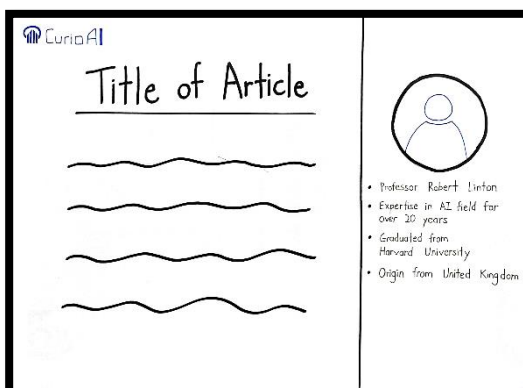
User could sign up or sign in



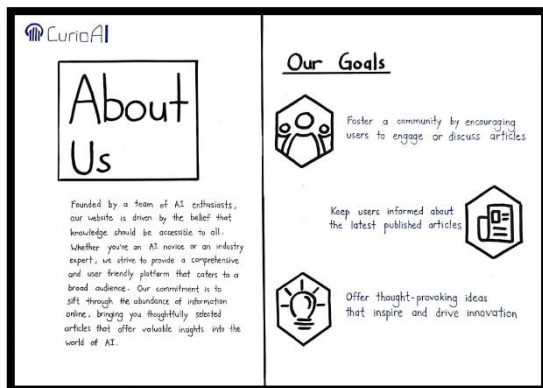
After creating account, user could choose their field of interests



The main page will be refreshed, and contents will be updated according to user's interests



When user select an article, it will show the bibliography of the author of the article at the right-hand side



User could select About Us to see our website history, goals, as well as contact number and email

Reflection

Brendan Chia Yan Fei



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Distribution of Tasks

Name		Lau Yan Kai	Neo Li Xin	Lee Yin Shen	Brendan Chia Yan Fei
Task	Interview and Documentation	Selecting AI expert to be interview			
		Request permission from expert	Video recording for evidence and record		
		Interview question preparation			
		Interview with expert			
		Documentation after interview			
	Prototype	Prototype testing & finalisation	Producing prototype	Video editing	
	Report Writing	Details description for each phase, Teamworking, Assessment point	Problem and Solution	Graphics and visual	Introduction
		Design thinking evidence			
		Reflection			
		Final review, finalisation			
		Presentation			

References